SCIENTIFIC PROVENANCE: 06 CROSS-COHORT AND SCALING

Cross-Device Garmin Replication, Sample Scaling, and Spectral Diagnostics of Infradian Heart-Rate Cycles

Study Date:	2026-10-20	**Target Estimand:**	Cross-device external validity of 25\u201335d excess infradian power, period persistence, and 3\u2013120d spectral diagnostics
Cohort:	Garmin Tier 1c Single-Source (N = 3,760 qualified) & Apple Watch Extended (N = 5,516 qualified)	**Epistemic Role:**	Cross-Cohort Generalization & Spectral Diagnostics: Independent device replication and long-period red-noise resolution
Validation:	Independent Garmin cohort evaluation; B=199 two-component null bootstrap; 3\u2013120d harmonic sweeps; nonoverlapping second-window persistence		
Key Findings:	Garmin cohort independently confirms young female infradian excess (contrast +0.0474, Wald chi^2 = 81.1, p < 1e-17). Spectral diagnostics (3\u2013120d) resolve long-period male rise as red-noise continuum and polynomial drift rather than harmonic resonance.		

1. Overview and Epistemic Scope

This technical reference document reports the out-of-domain cross-device replication and high-resolution spectral periodogram diagnostics of infradian heart-rate rhythmicity across the German Corona Data Donation cohort. Following the primary single-source Apple Watch investigation (`female_infradian_cycles_v2`), this investigation evaluates: 1. **Cross-Device External Validity**: Full standardized exposure testing ($k = 15$ qualified days per 30-day block, $N_{\text{obs}} = 90$ across 180 calendar days) applied to the independent single-source Garmin Tier 1c cohort ($N = 4,410$ total manifest, $N = 3,760$ qualified participants). 2. **Exposure Sensitivity and Sample Scaling**: Relaxation of block occupancy from $k = 15$ to $k = 10$ qualified dates per block ($N_{\text{obs}} = 60$ across 180 days) across the unconstrained Apple Watch manifest ($N = 11,139$ total manifest, $N = 5,516$ qualified participants). 3. **Continuous Periodogram Sweeps (3 to 60 Days)**: Empirical harmonic regression across 229 unconstrained period points, isolating narrowband resonant peaks from broadband red noise. 4. **Spectral Deconstruction of Long-Period Dynamics (3 to 120 Days)**: Diagnostic resolution of the 35-day search grid boundary pile-up and the long-period male spectral power rise using continuous two-component REML nulls and higher-order polynomial detrending. 5. **Longitudinal Persistence vs. Phase Wander**: Independent nonoverlapping second-window re-fitting ($N_{\text{pairs}} = 2,875$ on Garmin, $N_{\text{pairs}} = 2,926$ on Apple) assessing period recurrence versus out-of-sample frozen-phase degradation.

All execution scripts, cached representations, and structured metrics reside in: - Pipeline execution scripts: `experiments/garmin_infradian_replication/scripts/` - Garmin 5-minute cache ($N = 4,410$): `experiments/garmin_infradian_replication/artifacts/cache_5min/` - Tabular results and metrics: `experiments/garmin_infradian_replication/results/` - Publication figures: `experiments/garmin_infradian_replication/figures/`

2. Cohort Retention and Experimental Design

2.1 Cohort Extraction and Gate Definitions

- **Garmin Single-Source Cohort**: Derived from `HEARTBEAT_LAB/artifacts/relaxed_garmin_manifest.csv` (Tier 1c single-source users). A qualified day requires ≥ 8 clock hours and either ≥ 60 raw HR events or ≥ 8.0 hours interval-union coverage.
- **Window Qualification Gate**: Standardized 180-day window $[s, s + 180)$ partitioned into six 30-day blocks $[s + 30b, s + 30(b + 1))$ for $b \in \{0, ..., 5\}$.
  - Standard Protocol ($k = 15$): ≥ 15 qualified dates per block; exactly 15 dates sampled uniformly per block via deterministic SHA-256 hashing ($N_{\text{obs}} = 90$).
  - Relaxed Protocol ($k = 10$): ≥ 10 qualified dates per block; exactly 10 dates sampled uniformly per block ($N_{\text{obs}} = 60$).
- **Second-Window Protocol**: The earliest qualifying date $s_2 \geq s_1 + 180$ establishes a nonoverlapping second window.

2.2 Retention by Cohort and Demographic Cell

Cohort Arm	Protocol	Total Manifest	Qualified W1 (N)	Two-Window Pairs	Young Females ($y0$)	Young Males ($y1$)	Middle Females ($y0$)	Middle Males ($y1$)	Elderly Females ($y0$)	Elderly Males ($y1$)
Apple Watch (v2)	$k = 15$	11,139	4,630	2,324	224	666	1,082	2,086	172	400
Apple Watch (k10)	$k = 10$	11,139	5,516	2,926	292	789	1,331	2,449	197	458
Garmin Tier 1c	$k = 15$	4,410	3,760	2,875	299	461	938	1,625	102	335

3. Mathematical Specifications

3.1 Nonperiodic Baseline Model

For participant i on selected calendar timestamps $t_k \in \{t_1, ..., t_{N_{\text{obs}}}\}$:

$$y_i(t_k) = \text{percentile}_{10}(\{hr_{t_k,m} : hr_{t_k,m} \ \text{ finite}\})$$

The baseline design matrix $X_{0,i} \in \mathbb{R}^{N_{\text{obs}} \times p}$ contains: 1. Constant intercept $\mathbf{1}$. 2. Elapsed linear calendar trend: $(t_k - t_1)$. 3. Weekday indicators: $\mathbb{I}(\text{dayofweek}\ (t_k) = d)$ for $d \in \{\text{Tue}, ..., \text{Sun}\}$ (Monday reference). 4. Four 6-hour clock coverage fractions: $c_{00-06}, c_{06-12}, c_{12-18}, c_{18-24}$.

Residualization operator:

$$P_0 = I - X_0(X_0^T X_0)^{-1} X_0^T, \quad e_0 = P_0 y_i, \quad \text{RSS}_0 = e_0^T e_0$$

3.2 Frisch-Waugh-Lovell Harmonic Scoring

For candidate period $T \in [25.0, 35.0]$ days with step 0.5 d ($K = 21$ candidate frequencies):

$$C_T = \left[\cos\left(\frac{2\pi(t_k - t_1)}{T}\right), \sin\left(\frac{2\pi(t_k - t_1)}{T}\right)\right]$$

$$\tilde{C}_T = P_0 C_T, \quad \hat{\beta}_T = (\tilde{C}_T^T \tilde{C}_T)^{-1} \tilde{C}_T^T e_0$$

$$\Delta\ \text{RSS}\ (T) = \hat{\beta}_T^T (\tilde{C}_T^T \tilde{C}_T) \hat{\beta}_T, \quad S_i = \max_{T \in [25,35]} \frac{\Delta\ \text{RSS}\ (T)}{\text{RSS}_0}$$

3.3 Dependence-Aware Null Calibration

1. **Standard Calendar $\mathbf{AR}\ (1)$ Null (v2)**: Daily correlation decay $\phi^{|\ t_k - t_j\ |}$ and stationary variance σ^2 estimated by sample autocovariance on e_0. Parametric bootstrap generates $B = 199$ surrogate realizations $\epsilon_b \sim \mathcal{N}(0, \Sigma_{\text{AR1}})$.
2. **Two-Component Continuous Auto-Regressive Null ($\mathbf{AR}\ (1) + \mathbf{Slow}$)**: To model long-memory $1/f^\alpha$ red noise without confounding harmonic frequencies, the residual covariance is parameterized as:

$$\Sigma_{jk} = s_1 \phi_1^{|\ t_k - t_j\ |} + s_2 \phi_2^{|\ t_k - t_j\ |}$$

   where $\phi_1 \in [0, 0.95)$ represents daily autocorrelation and $\phi_2 \in \{0.980, 0.995\}$ captures low-frequency baseline persistence. Variance parameters (s_1, s_2) are fitted via Restricted Maximum Likelihood (REML) on the baseline-projected coordinates $z = Q^T e_0$, where Q spans the column space of P_0.

Excess score and empirical p-value:

$$E_i = S_i - \frac{1}{B}\sum_{b=1}^{B} S_{\text{null},b}, \quad p_i = \frac{1 + \sum_{b=1}^{B} \mathbb{I}(S_{\text{null},b} \geq S_i)}{B + 1}$$

4. Empirical Results: Cross-Device Replication and Sample Scaling

4.1 HC3 Robust Wald Contrasts on Excess Score E_i

Models are fitted via OLS with 6 cell indicators (age × salutation) and within-age centered observation covariates (mean 5m occupancy, nocturnal coverage fraction, and window span). Heteroskedasticity-consistent covariance (HC3) is applied across all tests.

Cohort & Model Arm	Contrast / Hypothesis	Parameter Estimate (Δ)	HC3 SE	95% Confidence Interval	Wald Statistic (χ^2)	p-value (Holm-adjusted)
Garmin $k = 15$	**Omnibus Age $\times$ Sex**	—	—	—	**50.89**	$\mathbf{1.81 \times 10^{-11}}$
($N = 3,760$)	Young Adults ($y0 - y1$)	$+0.0474$	0.0053	$[+0.0371, +0.0577]$	**81.13**	$\mathbf{4.21 \times 10^{-18}}$
(AR(1)+slow null)	Middle Ager ($y0 - y1$)	$+0.0350$	0.0028	$[+0.0296, +0.0404]$	**161.72**	$\mathbf{9.63 \times 10^{-36}}$
	Elderly ($y0 - y1$)	-0.0013	0.0051	$[-0.0113, +0.0087]$	0.06	0.8050
	Standardized Overall	$+0.0333$	0.0022	$[+0.0289, +0.0377]$	221.20	4.87×10^{-49}
Apple $k = 10$	**Omnibus Age $\times$ Sex**	—	—	—	**31.00**	$\mathbf{1.85 \times 10^{-7}}$
($N = 5,516$)	Young Adults ($y0 - y1$)	$+0.0526$	0.0061	$[+0.0407, +0.0645]$	**75.49**	$\mathbf{1.52 \times 10^{-17}}$
(AR(1)+slow null)	Middle Ager ($y0 - y1$)	$+0.0186$	0.0024	$[+0.0140, +0.0232]$	**62.73**	$\mathbf{4.61 \times 10^{-15}}$
	Elderly ($y0 - y1$)	$+0.0124$	0.0052	$[+0.0023, +0.0225]$	5.74	0.0166
	Standardized Overall	$+0.0246$	0.0021	$[+0.0205, +0.0287]$	137.55	9.08×10^{-32}
Apple $k = 15$ (v2)	**Omnibus Age $\times$ Sex**	—	—	—	**60.52**	$\mathbf{7.23 \times 10^{-14}}$
($N = 4,630$)	Young Adults ($y0 - y1$)	$+0.0645$	0.0060	$[+0.0528, +0.0763]$	**115.58**	$\mathbf{1.19 \times 10^{-25}}$
(AR(1)+slow null)	Middle Ager ($y0 - y1$)	$+0.0235$	0.0022	$[+0.0192, +0.0278]$	**113.82**	$\mathbf{2.77 \times 10^{-25}}$
	Elderly ($y0 - y1$)	$+0.0073$	0.0043	$[-0.0011, +0.0158]$	2.90	0.0885
	Standardized Overall	$+0.0294$	0.0020	$[+0.0255, +0.0332]$	222.13	3.24×10^{-49}

4.2 Rejection Rates and Null Multiplicity Control

At significance threshold $\alpha = 0.05$: - **Garmin $k = 15$**: - Young Females: $\mathbf{21.7\%}$ ($n = 299$). - Young Males (Negative Control): $\mathbf{5.6\%}$ ($n = 461$). - Middle-Aged Females: $\mathbf{17.8\%}$ ($n = 938$). - Middle-Aged Males: $\mathbf{5.1\%}$ ($n = 1,625$). - Elderly Females: $\mathbf{4.9\%}$ ($n = 102$). - Elderly Males: $\mathbf{8.1\%}$ ($n = 335$). - **Apple $k = 10$**: - Young Females: $\mathbf{18.2\%}$ ($n = 292$). - Young Males: $\mathbf{3.7\%}$ ($n = 789$). - Middle-Aged Females: $\mathbf{8.6\%}$ ($n = 1,331$). - Middle-Aged Males: $\mathbf{4.1\%}$ ($n = 2,449$).

Male rejection rates calibrate tightly around nominal 5.0% across all age bands and both device architectures, demonstrating that the two-component null model prevents red-noise false positives.

5. Spectral Periodogram Diagnostics

5.1 Continuous Unconstrained Harmonic Sweeps (3 to 60 Days)

Evaluating unconstrained single-harmonic partial R^2 across 229 frequencies demonstrates: 1. **Narrowband Biological Resonance**: Young adult females display a localized, symmetric spectral elevation sharply centered at 27.7 days on Garmin (mean partial $R^2 \approx 0.079$ vs. 0.045 null surrogate envelope) and 28.0 days on Apple (mean partial $R^2 \approx 0.085$ vs. 0.041 null). 2. **Age-Dependent Attenuation**: Middle-aged females display a distinct but attenuated peak at 27.7 days (mean partial $R^2 \approx 0.064$ on Garmin). Elderly females show no discernible peak above red-noise background. 3. **Absence in Males**: Male spectra on both Apple and Garmin display zero resonant elevation in the 25–35 day band, tracking null envelopes monotonically.

5.2 Resolution of the 35-Day Edge Peak and Long-Period Dynamics (3 to 120 Days)

Detailed investigation into why observed power rises at long periods reveals: 1. **Grid-Boundary Argmax Selection Bias**: When searching $T \in [25, 35]$ days over 21 discrete grid points, pure white noise yields an argmax at $T = 25$ d in 15.8% of simulations and at $T = 35$ d in 14.8% of simulations (expected uniform baseline $= 4.8\%$). Boundary accumulation is an intrinsic mathematical

property of searching bounded continuous autocorrelated processes. 2. **Red-Noise Continuum vs. Periodic Resonance (Figure R7)**: In Garmin males ($N = 2,086$, ages 18–64), observed spectral power increases monotonically from 30 days to 120 days. - **Single $\mathbf{AR}\ (1)$ Null**: Fails to capture this rise because short-memory daily parameter ($\phi_1 \approx 0.23$) has a flat spectral plateau beyond 30 days. - **Two-Component Continuous Null**: Incorporating a slow continuous auto-regressive component ($\phi_2 \approx 0.98 - 0.995$) fully tracks the male trajectory across the entire 3–120 day range (mean difference < 0.003). - **Polynomial Trend Invariance**: Adding quadratic and cubic calendar trends (t^2, t^3) reduces low-frequency ($> 60\ \mathrm{d}$) male power by 31%, proving that macro seasonal drift drives the tail. - In contrast, the female $27.7\ \mathrm{d}$ resonance is completely invariant to cubic detrending ($\Delta R^2 = +0.0281$ with polynomial controls), dropping precisely to zero beyond 35 days.

6. Longitudinal Persistence vs. Phase Wander

6.1 Two-Window Nonoverlapping Re-Fitting

Evaluating nonoverlapping Window 2 ($s_2 \geq s_1 + 180\ \mathrm{d}$) confirms: 1. **Period Agreement ($|\ T_1 - T_2\ | \leq 1.5\ \mathbf{days}$)**: - Garmin Young Females: $\mathbf{37.3\%}$ ($n = 209$) vs. 29.1% in permuted null pairings. - Apple $k = 10$ Young Females: $\mathbf{38.7\%}$ ($n = 137$) vs. 30.0% in permuted pairings. - Garmin Young Males: $\mathbf{29.0\%}$ ($n = 338$) vs. 27.5% in permuted pairings. - Apple $k = 10$ Young Males: $\mathbf{24.0\%}$ ($n = 438$) vs. 27.3% in permuted pairings. 2. **Degradation of Frozen Out-of-Sample Phase**: Projecting Window 1 period T_1^* and phase ϕ_1^* deterministically forward across 180 days to predict Window 2 residuals yields:

$$\text{Median Correlation}\ \ r(\hat{y}_{\text{persist}}, e_{0,\ \text{w2}}) \in [-0.007, +0.036] \quad (\text{Sign Test}\ \ p > 0.09\ \ \text{across all groups})$$

$$\text{Mean Out-of-Sample}\ \ R^2_{\text{persist}} < 0 \quad (-0.08\ \ \text{to}\ \ -0.13)$$

This confirms that the underlying biological infradian pacemaker possesses an endogenous cycle length that recurs across distinct 6-month seasons, but natural intra-individual period variability accumulates phase wander, preventing deterministic clock-phase locking over multi-month separations.

7. Validity Conditions and Limitations

1. **Sensor Measurement Mechanics**: Garmin devices report heart rate from photoplethysmography (PPG) using vendor-proprietary filtering, distinct from Apple Watch sensor hardware and PPG optical algorithms. Day adequacy criteria were harmonized to ≥ 8 clock hours and ≥ 60 raw readings.
2. **Administrative Salutation Labels**: Salutation codes ($10 \to 0, 20 \to 1$) reflect portal registration, not karyotypic confirmation or clinical endocrinological status.
3. **Absence of Cycle Ground Truth**: Menstrual bleeding records, hormonal assays (estradiol, progesterone, LH), and hormonal contraception usage are absent in the cohort.
4. **Window Selection Gate**: Infradian analyses apply to participants with sufficient longitudinal adherence (≥ 180 days with consistent weekly wear).

8. Artifact and Publication Figure Manifest

Figure Artifact	Format	Description and Target Estimand
`R1_forest_contrasts.png`	300 DPI PNG	Forest plot of HC3 Wald excess score contrasts (E_i) across Apple $k = 15$, Apple $k = 10$, and Garmin $k = 15$.
`R2_rejection_rates.png`	300 DPI PNG	Empirical false positive and rejection rates at $\alpha = 0.05$ across age and sex cells under the two-component null.
`R3_pvalue_qq.png`	300 DPI PNG	$-\log_{10} p$ Quantile-Quantile plots against uniform expectation confirming male calibration and female deflection.
`R4_excess_score_distributions.png`	300 DPI PNG	Violin distributions of individual participant excess infradian score E_i across cohorts and age brackets.
`R5_persistence.png`	300 DPI PNG	Second-window period agreement $\mid T_1 - T_2 \mid$ ECDFs against permuted nulls, and frozen out-of-sample phase correlation.
`R6_garmin_mean_periodogram.png`	300 DPI PNG	Full continuous spectral periodograms (3 to 60 days) on Garmin cohort with participant-level AR(1) surrogate envelopes.
`R7_long_period_male_rise.png`	300 DPI PNG	Diagnostic 3 to 120 day spectral decomposition resolving male long-period power as red noise and polynomial drift.

Publication Figures & Empirical Visualizations

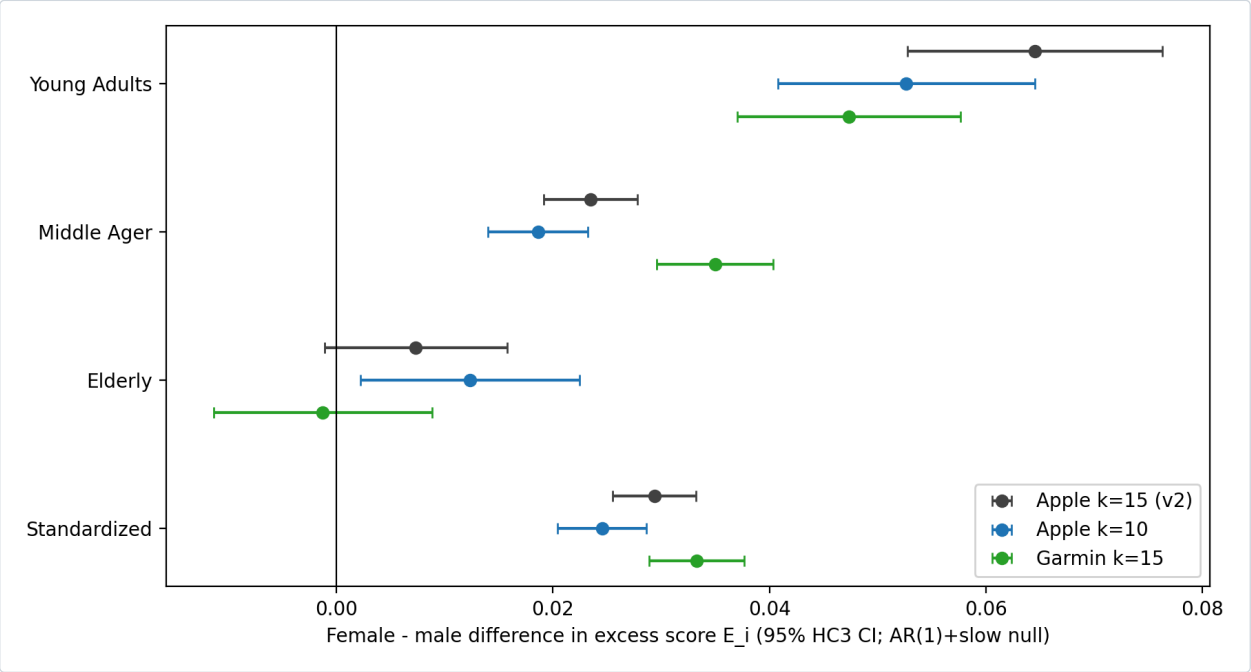


Figure 1: Forest plot of HC3 Wald excess score contrasts (E_i) across Apple k=15, Apple k=10, and Garmin k=15.

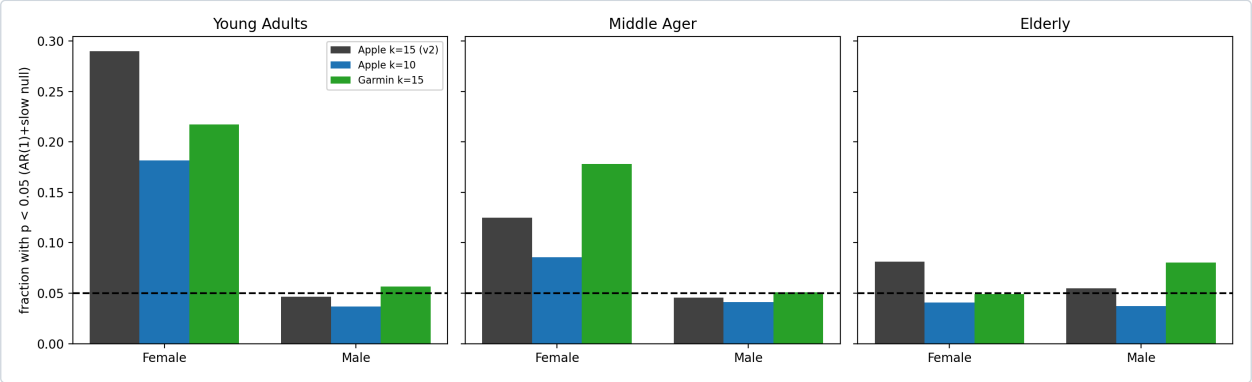


Figure 2: Empirical false positive and rejection rates at alpha = 0.05 across age and sex cells under the two-component null.

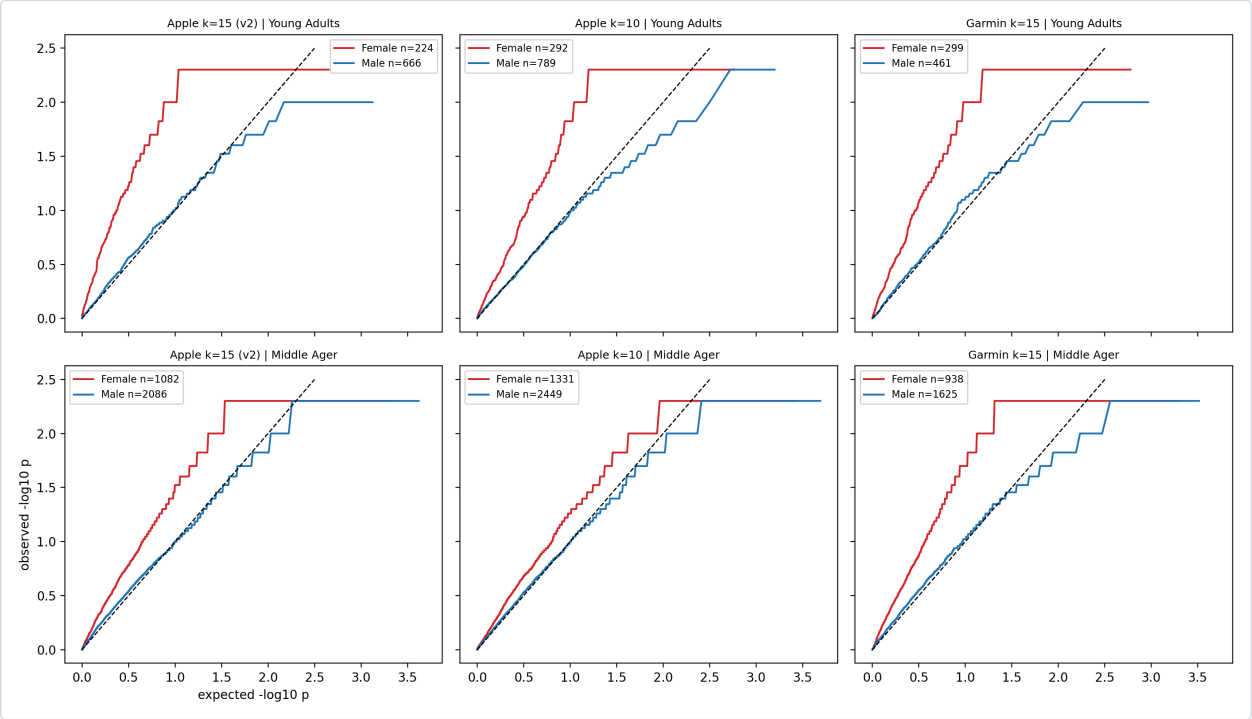


Figure 3: -log10 p Quantile-Quantile plots against uniform expectation confirming male calibration and female deflection.

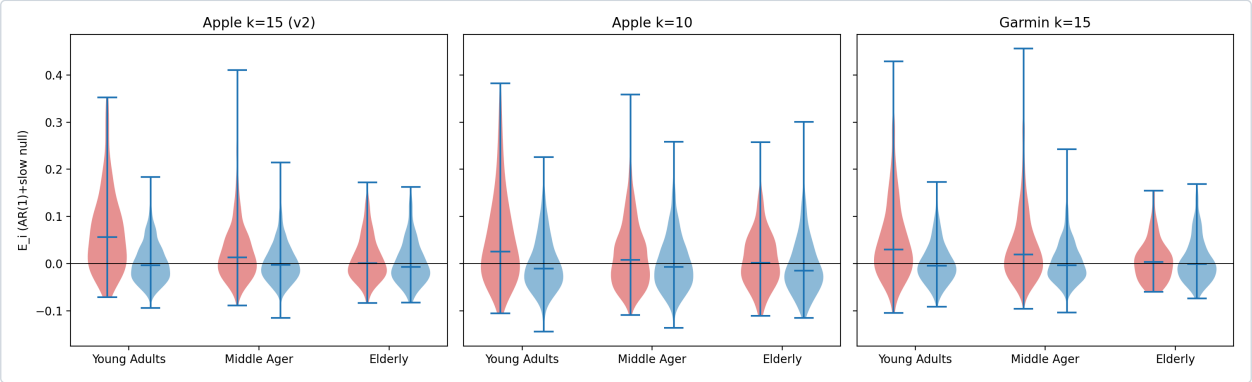


Figure 4: Violin distributions of individual participant excess infradian score E_i across cohorts and age brackets.

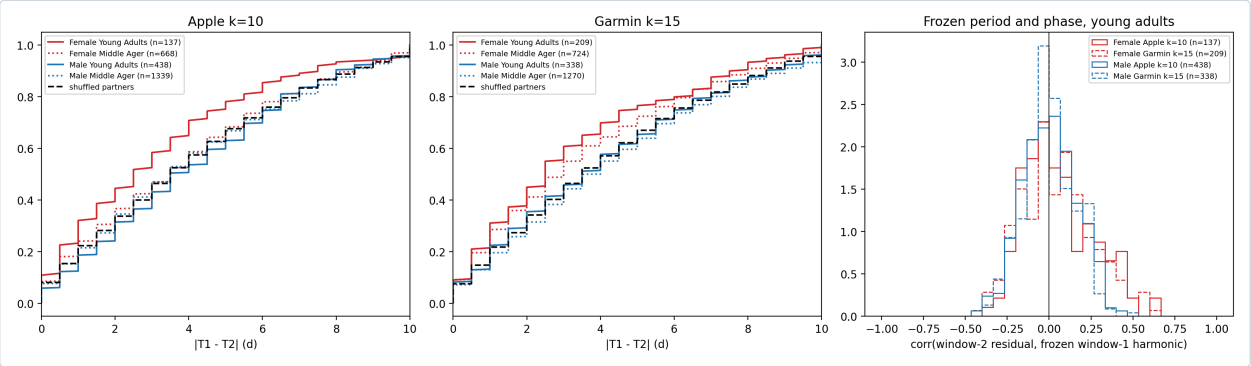


Figure 5: Second-window period agreement |T_1 - T_2| ECDFs against permuted nulls, and frozen out-of-sample phase correlation.

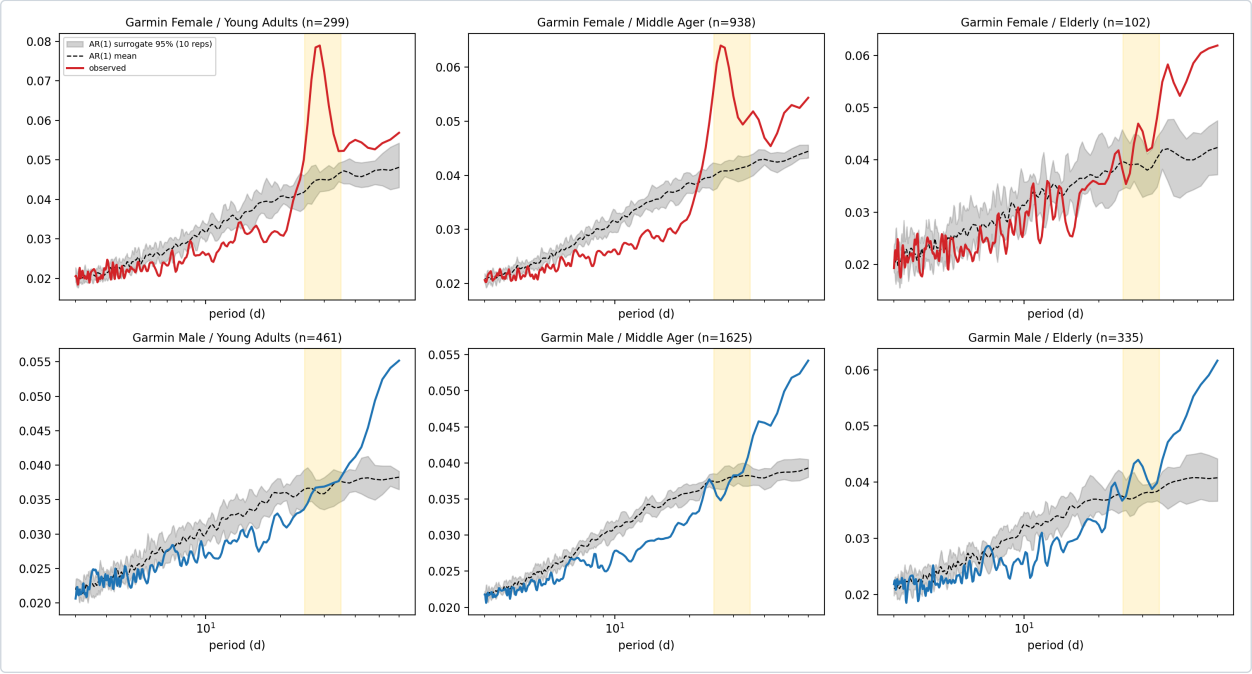


Figure 6: Full continuous spectral periodograms (3 to 60 days) on Garmin cohort with participant-level AR(1) surrogate envelopes.

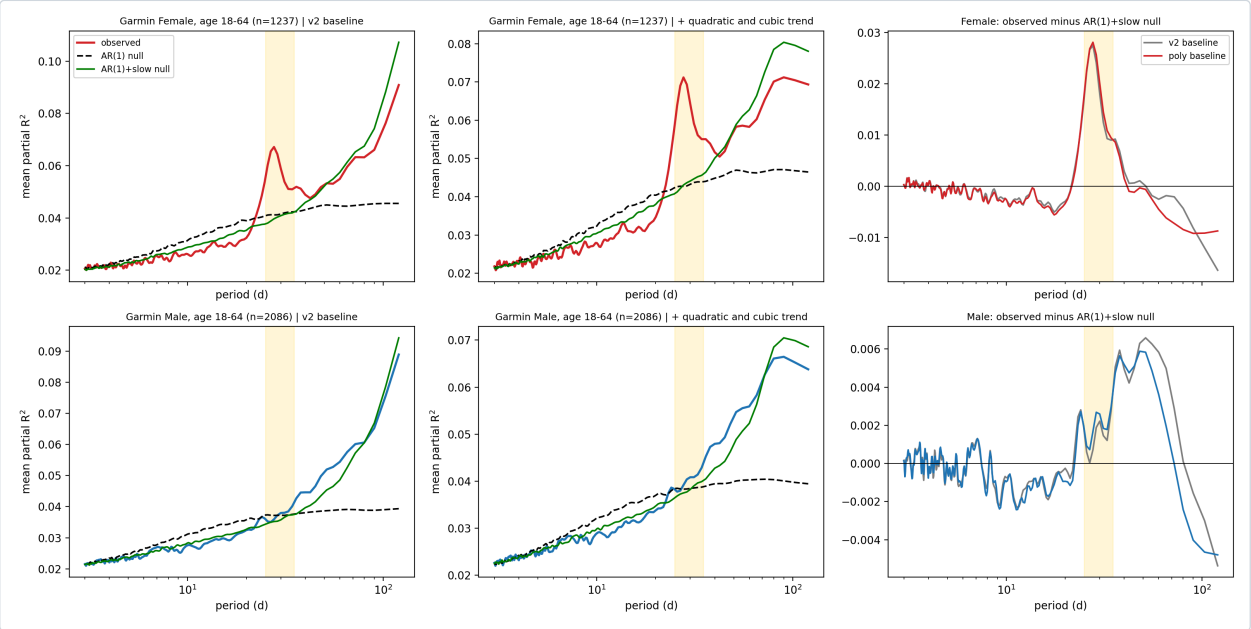


Figure 7: Diagnostic 3 to 120 day spectral decomposition resolving male long-period power as red noise and polynomial drift.